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The Importance of Cosmetic Stay-Green in Specialty Crops

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ABSTRACT

Stay-green plants are defined as those whose leaves have an extended period of green color compared to normal genotypes. Stay-green mutations are classified as either cosmetic or functional. Plants possess a functional stay-green phenotype if they continue to photosynthesize for a longer period than is normal. During the senescence of plants with cosmetic stay-green, chlorophyll is retained but the plant loses its capacity to photosynthesize. Cosmetic stay-green comes about as a result of mutations rendering enzymes in the chlorophyll catabolic pathway nonfunctional. Cosmetic stay-green mutants were first identified in forage/turf grass species and rice, and have since been identified in tomato, pepper, and common bean. In nearly every case, a homolog of the stay-green gene (*SGR*) is involved. Plant breeders have capitalized on the use of functional stay-green in field crops because of their ability to enhance yield and provide tolerance to abiotic stresses. Maize and sorghum breeders in particular have used functional stay-green to boost yields. On the other hand, cosmetic stay-green traits have been instrumental helping researchers gain a better understanding of the photosynthetic process, and especially the chloroplast catabolism pathway, but the trait has not been regarded as useful for genetic improvement. In this chapter, we wish to highlight the importance of cosmetic stay-green for quality traits in specialty crops. Genomics and bioinformatics tools, along with the sequenced genomes of a number of crops, have supercharged the study of senescence such that our understanding of the chlorophyll catabolic pathway is fairly complete and has enhanced the ability to associate genes with phenotypes. Several crops show phenotypic characteristics of stay-green traits, but they have not yet been characterized at the molecular level in

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order to confirm a mutation in the chlorophyll senescence pathway; these represent opportunities for further research. Such crops include cucumber, lentil, lima bean, mung bean, cowpea, and fava bean, among others. Cosmetic stay-green mutations contribute to increased quality in a number of crops, including dry edible peas and beans, garden peas and snap beans, tomatoes, and peppers. In some cases (e.g., peas), the economic value of cosmetic stay-green is considerable. Overall, the value of the stay-green characteristic in these crops has a positive economic impact all along the supply chain, from the producer to fresh markets and processing industries, and has the potential to benefit other crops. Cosmetic stay-green may also be associated with undesirable effects—mainly in relation to germination and vigor in certain species. This too represents a research opportunity.

KEYWORDS: chlorophyll, chlorophyll catabolism pathway, chlorophyll degradation, cosmetic stay-green, functional stay-green, green cotyledon, green seed, photosynthesis, senescence, SGR, stay-green

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ABBREVIATIONS

ACD1	Accelerated cell death 1
BAC	Bacterial artificial chromosome
BLAST	Basic Local Alignment Search Tool
CaSGR	*Capsicum annuum STAY-GREEN*
CBR	Chlorophyll B reductase
Chl a	Chlorophyll a
Chl b	Chlorophyll b
Chlide a	Chlorophyllide a
CHLASE	Chlorophyllase
DFCCs	Dioxobilin-type fluorescent chlorophyll catabolites
DNCCs	Dioxobilin-type nonfluorescent chlorophyll catabolites
FAO	U.S. Food and Agriculture Organization
GmSGR	*Glycine max STAY-GREEN*
Hydroxy-pFCC	Hydroxy-primary fluorescent chlorophyll catabolite
INDEL	Insertion/deletion
Mg	Magnesium
Mg-DeCHE	Mg-dechelatase
MtSGR	*Medicago truncatula STAY-GREEN*
NCCs	Nonfluorescent chlorophyll catabolites
ND	Not determined
NOL	NYC1-like
NYC1	Non-yellow coloring 1
NYE	Non-yellowing
OsSGR	*Oryza sativa STAY-GREEN*
PaO	Pheophorbide A oxygenase
PCR	Polymerase chain reaction
pFCC	Primary fluorescent chlorophyll catabolite
Pheide a	Pheophorbide a
Phein a	Pheophytin a
PPH	Pheophytinase
PsSGR	*Pisum sativum STAY-GREEN*
PvSGR	*Phaseolus vulgaris STAY-GREEN*
R2R3-MYB	Repeat 2 Repeat 3—myeloblastosis transcription factor
RCC	Red chlorophyll catabolite
RCCR	Red chlorophyll catabolite reductase
RNAi	RNA interference
RT-PCR	Reverse transcriptase polymerase chain reaction
RuBisCO	Ribulose-1,5-bisphosphate carboxylase/oxygenase
SGR	*STAY-GREEN*
SlSGR1	*Solanum lycopersicum* stay-green 1
SNP	Single-nucleotide polymorphism
THF1	Thylakoid formation 1

I. INTRODUCTION

This chapter came about from the realization that, as we have worked with various vegetable crops, delayed senescence has been of critical importance to plant improvement. Plant breeders have capitalized on the use of functional stay-green in field crops because of its ability to enhance yield and provide tolerance to abiotic stresses. On the other hand, cosmetic stay-green traits have largely been studied in order to better understand the photosynthetic process—in particular, chloroplast catabolism. Cosmetic stay-green is important for quality traits in many specialty crops, and it is our intent to collect existing information in the literature and call attention to potential new areas of research. Genomics and bioinformatics have given us the ability to associate genes with phenotypes. With these tools has come the awareness that apparently unrelated but economically important traits in a variety of vegetable and field crops are under the control of orthologous genes. In this chapter, we describe the stay-green trait, its function in photosynthesis, its phenotypic expression in various crops, and its potential and real economic impact.

A. Photosynthesis, Photoassimilate Storage and Remobilization, and Senescence

Green plants are so ubiquitous and abundant that we often forget how animals depend on them. Plants' ability to capture energy via sunlight and convert it into carbon profoundly altered the earth when oxygenic photosynthesis first arose ~2.3 billion years ago (Soo et al. 2017). Initially appearing in single-cell cyanobacteria and eukaryotes, the photosynthetic package was later coopted and incorporated into multicellular plants as the chloroplast. Chlorophyll, the most abundant pigment on the planet (Hörtensteiner 2009), is at the heart of the reaction that facilitates photosynthesis. Approximately 1 billion MT of chlorophyll is synthesized and catabolized on the planet each year (Kräutler 2008).

Chloroplasts are synthesized and chlorophyll accumulates after a seed germinates and the plant begins to grow. Chloroplasts continue to multiply within cells, and when cells undergo mitosis, both daughter cells receive a portion. The number varies from 10 to 200 chloroplasts per cell, depending on the cell type and the stage of growth and development (Possingham 1980). When fruits ripen or the plants senesce, chloroplasts and their contents are recycled and reused. Chloroplasts contain up to 75% of the nitrogen content in photosynthetically active cells,

and the most abundant protein in these organelles, RuBisCO, accounts for 20–30% of total nitrogen. Another large source of nitrogen in the chloroplast is found in the protein complexes of the thylakoids (Christ and Hörtensteiner 2014). Chloroplasts are also significant sources of metal ions (mainly Mg). During senescence, the turnover in chloroplasts and the orderly remobilization of nutrients is critical to optimal plant growth and to the development of storage organs such as seeds (Chen et al. 2016).

Photosynthesis is a phytotoxic process and is kept well compartmentalized within the cell. Chlorophyll acquires high-energy electrons from light—energy that, if not carefully dissipated, will damage organelles within the cell. To maintain cell viability, chloroplast catabolism is a carefully orchestrated process, proceeding through a number of steps as the molecule is disassembled and efficiently redistributed (Fig. 6.1). Termed the "PaO pathway," it contains two main parts. In the first, catabolism takes place in the thylakoid membranes upon the disassembly of the phototoxic catabolites. In the second, chlorophyll catabolites are rendered colorless and nonphototoxic, and are translocated from the chloroplast to the vacuole (Christ and Hörtensteiner 2014) (Fig. 6.1). A number of enzymes are involved in the catabolic process, and many of these are under nuclear genetic control (Ougham et al. 2008). Loss of function in one or more of these genes can alter photosynthetic processes and affect senescence. One of the main nonlethal phenotypic expressions of altered senescence is the stay-green phenotype.

B. What Is the Stay-Green Trait?

Stay-green traits alter the process of chlorophyll degradation. Stay-green mutants have been defined by Thomas and Smart (1993) as plants whose leaves have an extended period of green color compared to their parental genotypes. Stay-green mutations are classified as either cosmetic or functional. Plants possess a *functional* stay-green phenotype if they continue to photosynthesize for a longer period than is normal (Thomas and Smart 1993). Grain yields have been shown to increase up to 11% with only a 2-day delay in senescence (Thomas and Howarth 2000). Cultivars with functional stay-green may exhibit reduced drought stress, show decreased lodging, and have increased resistance to diseases (Jordan et al. 2003). Functional stay-green genotypes are known to store more carbohydrates in their stems (McBee and Miller 1982), assimilate more nitrogen, and have a greater specific leaf nitrogen content (Borrell et al. 2001; Richards 2006).

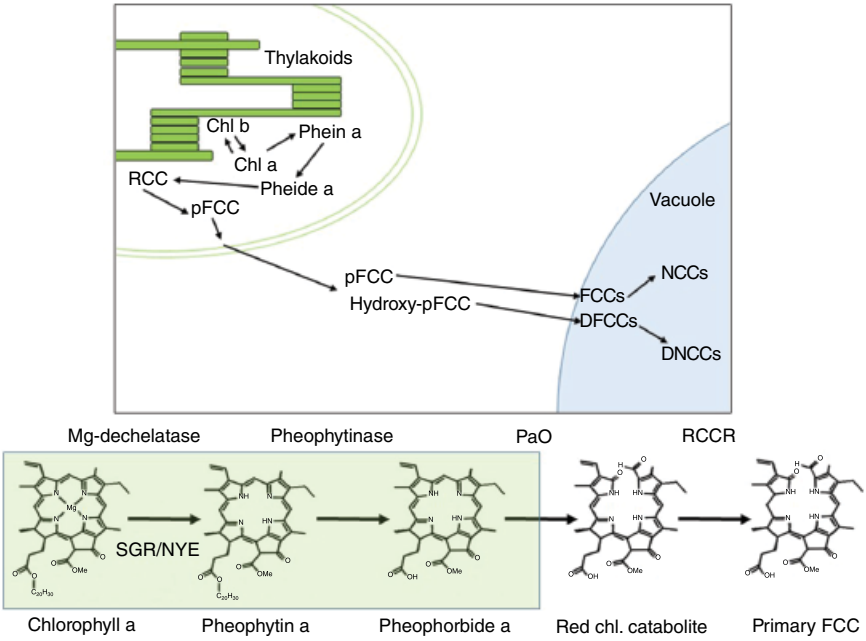

Fig. 6.1. The chlorophyll catabolic pathway. Top panel: The organellular location of chlorophyll and its derivatives, and their transport from the gerontoplast (senescing chloroplast) through the cytosol to the vacuole. Bottom panel: The chemical structure of chlorophyll a and its derivatives for the first four steps in the catabolic pathway. Compound names are shown below the molecules, and the enzymes responsible for each step are listed above. The shaded box over the first two steps in the catabolic pathway indicates that these molecules retain their green color. *Source:* Catabolic pathway figure modified and used with permission of Harpaz-Saad et al. (2007) and updated with information from Chen et al. (2016), Christ and Hörtensteiner (2014), Kuai et al. (2017), and Mach (2016).

The physiological basis of the functional stay-green trait is not well understood, and it is unclear whether carbon storage in stems and greater leaf nitrogen content are a consequence of stay-green mutations or are secondary traits associated with the general adoption of stay-green (Harris et al. 2006). Nonetheless, stay-green's potential for increased yields suggests the value of additional investigation. Functional stay-green mutants have been identified in rice, sorghum, wheat, and maize, and have been actively used in breeding for high yields and abiotic stress tolerance.

During the senescence of plants with c*osmetic* stay-green traits, chlorophyll is retained, although the plant loses its capacity to

photosynthesize (Thomas and Howarth 2000). Phenotypically, these plants remain green in color during senescence and as dry-down of the biomass takes place. Other phenotypic characteristics may include green seeds (cotyledons or testa) and accumulation of pheophytin a in the fruit of various crops.

II. CHARACTERISTICS OF COSMETIC STAY-GREEN

Thomas and Howarth (2000) have articulated a more nuanced view of functional and cosmetic stay-green, describing five stay-green syndromes based on their influence on chlorophyll during leaf senescence. Types A and B stay-green are regarded as functional due to prolonged photosynthetic activity during seed fill. Types C, D, and E stay-green are considered cosmetic; although chlorophyll is retained, photosynthetic activity compares to that of a normal plant in similar stages of senescence. Compared with the wild type, the induction of senescence is delayed in type A stay-green, although chlorophyll degrades at the same rate as in wild-type once senescence occurs. Type B mutants initiate senescence at the same time as wild types, but the degradation of chlorophyll is significantly slower. Type C stay-green retains chlorophyll almost indefinitely, although other aspects of senescence proceed normally. Normal chlorophyll degradation is disrupted in Type C. Type D stay-green results in leaf death by sudden freezing or drying. Finally, type E stay-green retains high chlorophyll content without increased photosynthesis.

While functional stay-green traits may improve yield and stress tolerance (Jordan et al. 2012), the benefit of cosmetic stay-green variants is not immediately obvious. Type C stay-green mutants in various crop species have been unconsciously selected for their novel appearance during growth and production, rather than for their prolonged greenness during senescence. In this chapter, we focus on type C cosmetic stay-green types, where chlorophyll catabolism is interrupted. Brown et al. (1991) classified leaf senescence reactions as either type I or type II. The enzymes chlorophyllase, Mg-dechelatase, and pheophorbide catalyze type I reactions and result in the opening of tetrapyrrole ring V (Fig. 6.1). The Type II reactions lead to cleavage and opening of the porphyrin macrocycle (Brown et al. 1991). Cosmetic type C stay-green variants belong predominantly to the type I senescence reaction group. This group has reduced pheophorbide a oxygenase (PaO) activity and stable pigment–protein complexes within the chloroplast (Thomas et al. 2002).

Table 6.1. Type C (cosmetic) stay-green mutants in crop and model plant species that have been characterized at the molecular level.

Species	Allele/GENE	Accession	Protein	Reference
Arabidopsis thaliana	*nye1-1/NYE1*		NYE1	Ren et al. (2007)
	ore 10		ND	Oh et al. (2003)
	ore 11		ND	Oh et al. (2003)
	NOL		CBR	Horie et al. (2009)
	pph-1/PPH		PPH	Schelbert et al. (2009), Morita et al. (2009)
	ACD1		PaO	Pružinská et al. (2003), Tanaka et al. (2003)
			THF1	Huang et al. (2013)
	SGR		Mg-DeCHE	Shimoda et al. (2016)
	NYC1		CBR	Kusaba et al. (2007)
Brassica napus	*NYE1*		ND	Qian et al. (2016)
Capsicum annuum	*cl*	Ancho	SGR	Barry et al. (2008), Borovsky and Paran (2008)
	cl?	Negral	ND	Roca and Mínguez-Mosquera (2006), Roca et al. (2006)
Citrus sinensis	*nan*		SGR	Alós et al. (2008)
Festuca pratensis	*Sid*	Bf 993	SGR	Vicentini et al. (1995), Thomas and Smart (1993)
Glycine max	*cytG*		ND	Canfield et al. (1995)
	d1d2		ND	Canfield et al. (1995)
Lolium (*Festuca* introgression)	*y*		SGR	Armstead et al. (2006), Armstead et al. (2007)
Medicago truncatula	*MtSGR*	NF2089	ND	Zhou et al. (2011)
Oryza sativa	*nyc1/NYC1*		CBR	Kusaba et al. (2007), Horie et al. (2009)
	sgr	Huazhiwu	ND	Jiang et al. (2007), Park et al. (2007), Sato et al. (2007)
Phaseolus vulgaris	*pc*	Alamo, Hystyle	SGR	Bachmann et al. (1994), Davis et al. (2009)
Pisum sativum	*i*	JI2775	SGR	Armstead et al. (2007), Sato et al. (2007)
Solanum lycopersicum	*gf*	LA3534	SGR	Barry et al. (2008), Barry and Pandey (2009)

Table based on Hörtensteiner (2009) and Kusaba et al. (2013), and updated with recent literature.

Type C stay-green mutations have been the most extensively studied, because they provide insight into chloroplast disassembly during senescence. This mutation was first discovered in Fescue (*Festuca pratensis*) forage grass, and was named *Sid* for *senescence-induced degradation* (Thomas and Stoddart 1975; Thomas 1987). In recent years, type C stay-green mutants have been identified in *Arabidopsis*, rice, and pea (Table 6.1), and have been generally designated as *SGR* or *NYE* (Jiang et al. 2007; Park et al. 2007; Balazadeh 2014). Stay-green mutants have since been identified in tomato, pepper, and common bean; these appear to be conditioned by *SGR*, as in other species (Table 6.1).

Great strides have been made in understanding chlorophyll catabolism at the molecular level, based on recent research in *Arabidopsis thaliana*. In this species, the gene family consists of three members: closely related *SGR1* and *SGR2*, and more distantly related *SGRL*, which lacks the C-terminal motif present in *SGR1* and *SGR2*. *SGR1* is also synonymous with *NYE1*, where the mutant allele *nye1-1* produces the stay-green phenotype (Balazadeh 2014). *SGR1* appears to be directly involved in catabolism of chlorophyll, because knockouts of this gene produce the stay-green phenotype, while its overexpression enhances rate of leaf yellowing. Shimoda et al. (2016) used a wheat-germ protein-expression system to provide evidence that *SGR1*/*NYE1* is an Mg-dechelatase. *SGR2*/*NYE2* acts as a negative regulator of *SGR1* and does not express a phenotype when *SGR1* is nonfunctional. Knockouts of *SGR2* result in normal senescence, whereas overexpression produces a stay-green phenotype. Both *SGR1* and *SGR2* are localized in the thylakoid membranes, where expression is minimal in presenescent tissues, but ramps up during senescence. The function of *SGRL* is less well understood. It shows high levels of expression in presenescent chloroplasts but decreases during senescence. Balazadeh (2014) suggested that *SGRL* may be involved in presenescence chlorophyll turnover. It shows a much stronger substrate affinity for Chlide a than do *SGR1* and *SGR2* for Chl a (Shimoda et al. 2016), and it converts chlide a directly to pheide a (Chen et al. 2016). Sakuraba et al. (2014) demonstrated that *SGRL* forms homo- and heterodymers with *SGR1* and *SGR2*, facilitating chloroplast catabolism during stress-induced senescence. The picture that emerges is that of a complex of six catabolic enzymes bound by *SGR* to the light harvesting complex II that facilitates the diassembly of chlorophyll and prevents phytotoxic derivatives from escaping to damage the cell.

In the remainder of this chapter, we will mainly discuss type C cosmetic stay-green variants. Unless otherwise noted, we will refer to the gene that confers the stay-green phenotype as *SGR*, with the

understanding that this most likely corresponds to the *SGR1*/*NYE1* of *A. thaliana*. Based on homology, this appears to be true for fescue, pea, rice, pepper tomato, and common bean, but as further detail emerges from the study of these species, it may be that other members of the *SGR* gene family—or even other genes in the catabolic pathway—are involved in different species. In Table 6.1, the gene (capitalized) is listed along with the allele (lower case) conferring the stay-green phenotype, where known. The "protein" column in Table 6.1 lists the enzyme, if known, for which the gene encodes. In some cases, the enzyme has not been identified, but given the high degree of homology with *Arabidopsis*, it is listed as "SGR." What follows is a summary of what is known of cosmetic stay-green mutations in various crops of economic importance.

III. GENOTYPIC AND PHENOTYPIC DESCRIPTION, AND ECONOMIC IMPORTANCE BY PLANT FAMILY

A. Poaceae

1. Fescue and Ryegrass. One of the first stay-green variants to be identified was the line Bf 986, discovered in Wales by Lewis (1972) in the forage grass *F. pratensis*. This material was initially characterized by Thomas and Stoddart (1975), who found that the mutant phenotype retained chlorophyll and green leaves even as the normal phenotype turned yellow and lost its chlorophyll. Further research revealed that it was a type C stay-green mutant, and it became the model species for learning much of what we know of chlorophyll catabolism (Armstead et al. 2006). The trait was transferred to perennial ryegrass (*Lolium* spp.), where *Sid* was mapped to chromosome 5 and was shown by synteny with rice to co-segregate with *SGR* (Armstead et al. 2006).

The stay-green trait, when used in perennial ryegrass, has economic value in forage and turf grasses (Armstead et al. 2006). Of special interest is the retention of color under stress conditions (Armstead et al. 2009). Little information is available on how widely the trait has been incorporated by breeders into forage and turf grass cultivars.

2. Rice. While rice is not a specialty crop, it was the first species for which cosmetic stay-green was characterized at the molecular level and for which a candidate gene was implicated. The mutant version of *SGR* was induced in Japonica rice (*Oryza sativa*) cv. 'Huazhiwu' by x-ray irradiation (Cha et al. 2002). It was denoted as *sgr(t)* and mapped to chromosome 9. Biochemical analysis revealed that this stay-green

mutant possessed three characteristics during senescence: retention of chlorophyll, stability of chlorophyll–protein complexes, and stability of thylakoid membrane structures, but with loss of the ability to photosynthesize. A separate recessive stay-green mutant, designated *sgr*, also mapped to chromosome 9, and had phenotypic characteristics similar to the 'Huazhiwu' mutant in that it delayed yellowing but without maintaining photosynthesis (Cha et al. 2002). Functional analysis of *SGR* mutations in rice has shown that the introduction of a rice *SGR* mutation under the control of the *OsSGR* promoter restores the *SGR* stay-green phenotype. By contrast, the introduction of a mutated *OsSGR* does not, because the insertion of an amino acid leads to defects in *SGR* function.

The *SGR* gene expresses in thylakoid membranes of the chloroplast in rice. Like *PaO*, *SGR* is upregulated by dark-induced senescence in rice leaves. When *SGR* is overexpressed, the number of lamellae in the grana thylakoids—and thus the chlorophyll content of normally growing leaves—falls. Chlorophyllide a oxygenase accumulates at reduced levels, as well (Cha et al. 2002).

Stay-green characteristics can be detected during vegetative growth and grain-filling. During a dark-induced period, leaves of the wild-type japonica rice 'Hwacheong-wx' turn yellowish-brown, while plants carrying an *SGR* mutation remain green much longer, due to the slower degradation of chlorophyll (Thomas and Howarth 2000). There are close similarities between *SGR* in rice and Bf993, the type C stay-green mutant of *F. pratensis* (Hauck et al. 1997; Thomas and Howarth 2000).

Although the cosmetic stay-green trait in rice has revealed details of chlorophyll senescence and the underlying molecular biology of this process, the trait does not appear currently to have economic value in this crop. If the trait functions similarly in the seed of rice to how it does in legumes, then *SGR* rice may have green seed color, and the altered color of the grain may have commercial value for its novel color.

B. Fabaceae

1. Pea. *Pisum sativum* was initially domesticated as a grain crop, from which the vegetable forms were derived. These include shell peas, where the mature but high-moisture seeds are eaten, and the low-fiber pods of snow and snap peas. Ancestors of dry pea had yellow cotyledons, but the green cotyledon trait appears to have arisen early on and was well established by the time Gregor Mendel conducted his experiments with garden pea (Fig. 6.2). Mendel observed segregation ratios for green and yellow cotyledon color (as well as six other

Fig. 6.2. Dry edible pea cotyledons prepared as split peas. Cotyledons at the top have the stay-green trait (*i*), while those below have the wild-type cotyledon color (*I*).

traits) over generations in order to formulate his laws of inheritance (Mendel 1866). Green cotyledons are recessive to yellow cotyledons, and the trait was given the gene symbol *i* (White 1917). In addition to green cotyledons, plants with *i* retain green leaves, stems, and pods as they senesce. Thomas et al. (1996) demonstrated that green cotyledon in pea is the result of a metabolic deficiency in the chlorophyll breakdown pathway that normally acts only during seed maturation. Similar to *SGR* in rice, *i* appears to be a type C stay-green mutation involved in the PaO pathway of chlorophyll catabolism. The green cotyledons exhibit loss of SGR protein and reduced expression of *SGR*. Prior to the identification of *SGR* as a candidate gene for *I*, researchers

had hypothesized that *PaO* was the candidate for *I*. Moffet and Weeden (2005) demonstrated that *PaO* was located on chromosome 7 and therefore could not be a candidate for *I*, which resides on chromosome 1. Armstead et al. (2007) showed co-segregation between the *I* locus and *PsSGR*, and demonstrated through transformation of an RNAi construct in *Arabidopsis* that *PsSGR* has a stay-green phenotype. Sato et al. (2007) used the rice *SGR* to identify a homolog in pea, and discovered three *i* alleles, two of which expressed a stay-green phenotype, while the third did not. Sequence analysis revealed that *PsSGR* consists of four exons separated by three introns. Sato et al. did not find a correlation between the cotyledon color phenotype and the intron-length polymorphism, and concluded that the stay-green phenotype is caused by defects in exons in *PsSGR*.

After soybean and common bean, pea represents the third-ranked grain legume in terms of economic importance. Dry pea had a global production of about 11.2 million MT in 2014 (FAOSTAT 2017). About 40% of dry grain-type peas have green cotyledons, and 60% yellow (K. McPhee, Montana State University, pers. comm.). Peas used as a vegetable are entirely green cotyledon, so the production of the crop—17.4 million MT—is the value of this stay-green trait in vegetable peas. The U.S. Food and Agriculture Organization (FAO) does not report a dollar value for production data, but in the United States the value of the 2016 green pea crop was US$95.7 million, and that of the dry pea crop US$298.2 million (USDA/NASS 2017).

2. Bean. Dry and snap beans (*Phaseolus vulgaris*) are the grain legume and vegetable forms of common bean, respectively. Dry beans have evolved into a number of market classes based primarily on seed color, among them the green-seeded French flageolet (or haricot vert) (Fig. 6.3). Snap beans appear to have been derived from dry beans in Europe following the Columbian exchange (Myers and Baggett 1999). From Europe, snap beans as well as some dry bean market classes were reintroduced to the Americas and elsewhere around the world.

Flageolet types have large, pale-green seeds (Fig. 6.3), which are eaten either as shell-outs (high-moisture but mature seeds) or as dry beans. Honma et al. (1968) examined inheritance in approximately 1700 F_2 plants and found a lack of fit to a 3 : 1 ratio, but a 13 : 3 ratio for segregation. They concluded that the green seed trait was controlled by two genes, a green factor (Gr) and an inhibitor (Ih), with individuals expressing green seed color in the Gr-ihih genotypes. An alternative explanation for the observed segregation ratio is that the trait is controlled by a single recessive gene but segregation ratios are skewed

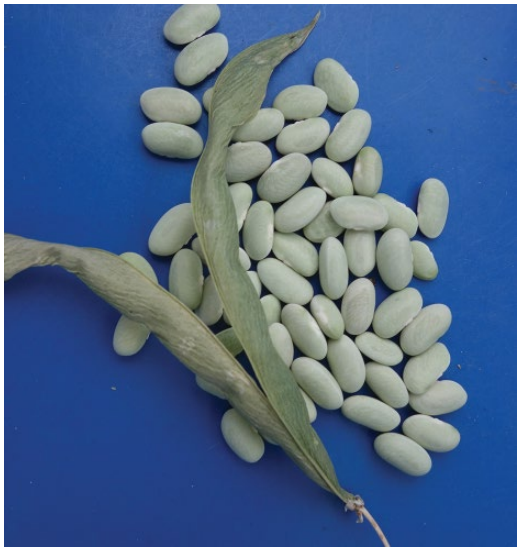

Fig. 6.3. Seeds and pods of the common bean *persistent color* (*pc*) cultivar 'Pascal,' a Flageolet type. Note the senesced and dry but green pod and the pale green seed.

towards a deficit of the green seed class. This does indeed turn out to be the case both in our own experience with the trait (see later) and as characterized by Dean (1968). In his studies, Dean determined that the trait was controlled by a single recessive gene (Mayland and Dean 1971). Various names for the trait are evident in these early publications, but when it was added to the list of common bean genes, it was assigned the name *persistent color* (*pc*) (Baggett et al. 1999). As a snap bean breeder, Dean (1968) was interested in incorporating *pc* into snap bean because it produces pods with a highly uniform green color. Major commercial breeding programs subsequently acquired and used Dean's germplasm to introgress the trait into commercial snap bean cultivars (Fig. 6.4).

Bouwkamp and Honma (1970) and Ronning et al. (1991) examined the physiological differences between *pc* and normal beans and determined that the persistent green color extended to the foliage. Leaf chlorophyll persisted, but with a loss of photosynthetic activity, indicating that *pc* behaves as a type C stay-green (Thomas and Smart 1993). Further characterization by Bachmann et al. (1994) of chlorophyll breakdown products supported the type C stay-green classification of this trait.

Davis et al. (2009) developed molecular markers to investigate whether the gene regulating the *pc* trait is a member of the *SGR* family. They

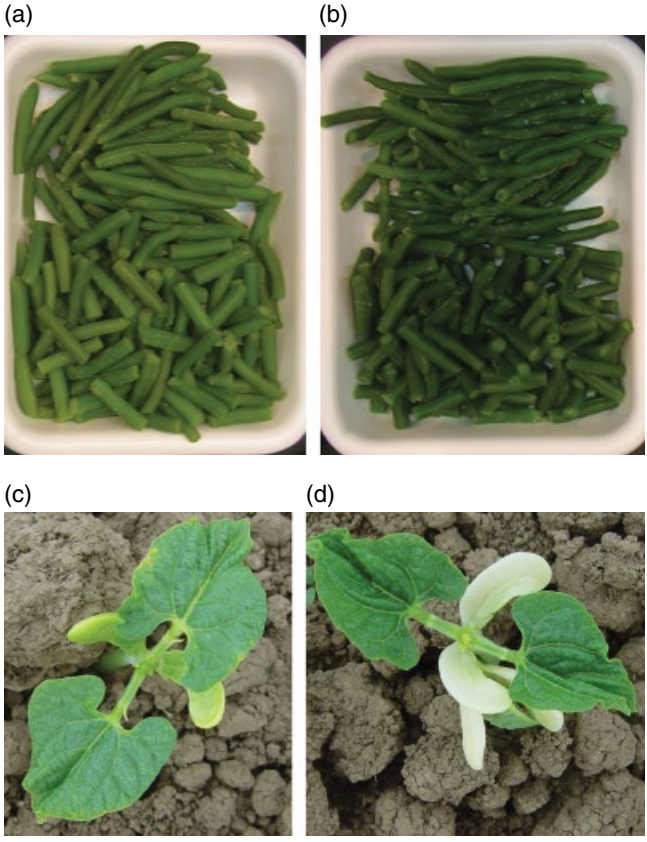

Fig. 6.4. Snap bean pods and seedlings, showing phenotypic differences between normal and *persistent color* (*pc*) lines. (a) Frozen processed pods of a normal snap bean line. (b) Frozen processed pods of a *pc* snap bean line. (c,d) Normal snap bean seedlings, showing (c) green and (d) white cotyledon color of *pc* seedlings.

Basic Local Alignment Search Tool (BLAST)-searched the *Phaseolus* genome using a *PsSGR* sequence (Sato et al. 2007) and obtained an orthologous sequence for common bean. They then designed two sets of primers: one that amplified a 1000 bp fragment in both normal and *pc* types, and one that used one primer from the previous set and a second further downstream, which detected a 1450 bp fragment in normal but not in *pc* genotypes. In three F_2 segregating populations (n = 215), the *pc* phenotype co-segregated with the markers. Based on these data, the authors concluded that an insertion/deletion (INDEL) in *PvSGR* is responsible for the mutant phenotype. The hypothesis that

pc is a candidate for *SGR* awaits confirmation by transgenic expression experiments.

The total production of dry bean worldwide in 2014 was about 26.5 million MT (FAOSTAT 2017). Snap bean is a widely consumed vegetable, with some 1.8 million MT produced worldwide in 2014 (FAOSTAT 2017). FAO statistics are somewhat problematic in obtaining accurate data on common bean, however (Myers and Kmiecik 2018). Species other than *P. vulgaris* may be grouped under various headings, and the FAO designation of "dry bean" may include other bean species (excluding cowpea). The FAO classification "green bean" consists of shell-out beans and may include various species. Snap beans are designated by the FAO as "string beans," which may also include *Vigna* species. Snap bean was valued at US$441.3 million in the United States in 2016 (USDA/NASS 2017), where some 40% of the acreage was planted to *pc* types (K. Kmiecik, pers. comm.).

3. Soybean *Glycine max* is known mainly as a source of vegetable oil and high-protein defatted meal that is used as an animal feed and processed for human consumption. Like other grain legumes, a vegetable form ("edamame" in Japanese, "mao dou" in Chinese) evolved in East Asia (Dong et al. 2014). Vegetable soybean cultivars have large, plump, flavorful seeds that are eaten at maturity but before senescence has commenced. Seed coats may vary in color but are mainly yellow or green (Mimura et al. 2007; Dong et al. 2014).

Soybean is an ancient polyploid, and as a result, about 75% of its genes have duplicate loci (Nakano et al. 2014). Terao (1918) and Woodworth (1921) identified variants for seed and cotyledon color in soybean. These included three type C stay-green variants that express in the leaves and seed (Thomas and Smart 1993). Two traits show Mendelian inheritance and are controlled by a single dominant (*G*) or a pair of recessive (*d1d2*) genes. *G* alone produces a stay-green phenotype in the seed testa but not the cotyledons, pods, and leaves. Plants with the combined *d1d2* duplicate loci exhibit a stay-green phenotype in the seed coat, cotyledons, pods, and vegetative portions (Thomas and Smart 1993). When *G* and *d1d2* are combined, plants exhibit a stay-green phenotype in all tissues. A third trait is controlled by the cytoplasmically inherited *cytG* gene. Plants with *cytG* exhibit the stay-green phenotype in leaves, pods, seed testa, and cotyledons (Thomas and Smart 1993).

Fang et al. (2014) and Nakano et al. (2014) concluded that *GmSGR1* (Glyma01g42390) and *GmSGR2* (Glyma11g02980) are candidate genes

for *d1* and *d2*. Nakano et al. (2014) discovered five *GmSGR* genes with *GmSGR1* and *GmSGR2* showing a high degree of similarity to *SGR* orthologs in *P. vulgaris* and *Vigna unguiculata*. *GmSGR3a*, *GmSGR3b*, and *GmSGR4* appear to be nonfunctional pseudogenes. Fang et al. (2014) found eight *SGR* gene models; the two already described were orthologous to *P. sativum* and *O. sativa SGRs*, and to *A. thaliana NYE1* and *NYE2*. Transformation experiments confirmed gene function in both cases.

By sequencing the chloroplast genome in normal and *cytG* backgrounds, Kohzuma et al. (2017) found a 5 bp insertion in *cytG* types, resulting in a frameshift mutation in a low-molecular-weight subunit of photosystem II. This gene apparently plays a duel role in photosynthesis and chlorophyll catabolism, and may interact with the nuclear gene *NYC1*. To date, a candidate gene for *G* has not been reported.

For commercial oilseed and protein grain-type soybeans, there is a clear preference for yellow seeds, and cosmetic stay-green may actually be a detriment because retained chlorophyll can interfere with oil quality (Smolikova et al. 2017). It would seem that there should be a preference for green seed coat and cotyledons for vegetable soybeans, at least at the seed stage preferred for consumption. Germplasm and genetic diversity surveys of vegetable soybeans have in fact found both green and yellow seed colors among accessions (Mimura et al. 2007; Dong et al. 2014). Kohzuma et al. (2017) surveyed a collection of 212 green cotyledon Asian soybean accessions (both vegetable and grain types) and found that those of Japanese origin possessed *ctyG*, those of Chinese origin had *d1d2*, and those of Korean origin had both. This finding suggests that Chinese and Japanese soybeans underwent independent selection for seed traits. In traditional cuisine, Koreans prefer soybean seed with a black seed coat and green cotyledon to that with a black coat and yellow cotyledon (Kang et al. 2010). In a collection of Chinese vegetable soybeans, approximately 57% were yellow- and 23% were green-seeded (the remaining 20% were black or brown in color) (Dong et al. 2014), whereas a collection that included Japanese, Chinese, and American accessions showed a more even split (54% green, 42% yellow, and 4% other colors) (Mimura et al. 2007). Considering the cultural preference for green cotyledons in Korea, and the substantial presence of green seed in vegetable soybean collections, it would seem that stay-green is a significant trait in this class. While worldwide production of soybeans (306.5 million MT in 2014) is the highest among any edible grain legume (FAOSTAT 2017), that of Korean traditional types and vegetable soybean is not known.

C. Solanaceae

1. Tomato. *Solanum lycopersicum* shows a high degree of diversity in fruit color, due to the different carotenoids present. Kerr described the green flesh mutation (*gf*) of tomato in his study of fruit color (Kerr 1956). He located the *gf* gene on chromosome 8 (Kerr 1957). Tomato fruit from plants with *gf* appear similar to normal tomato fruit when immature, but acquire a distinctive reddish-brown color when ripe in cultivars where the predominant carotenoid is lycopene and color in the epidermis is expressed (conditioned by *Y*) (Table 6.2, Fig. 6.5a,b). In plants with colorless epidermis (*y*) and lycopene, ripe fruit are pinkish-purple. Plants with *r* have a block in the carotenoid biosynthetic pathway that leads to the accumulation of phytoene, and express as pale-yellow when ripe. With the addition of *gf*, these fruits show a distinctive ripe green color (Table 6.2, Fig. 6.5c).

Early research characterized fruit maturation in tomato, but persistence of chlorophyll in the leaf was overlooked until the focus shifted to molecular characterization of tomato-leaf senescence ((McGlasson and Poovaiah 1975; Drake et al. 1996). As fruit ripening and leaf senescence proceed, a difference in chlorophyll content is observed between *gf* and the wild-type. *Green flesh* types have reduced chlorophyllase activity, and chlorophyllides a and b accumulate during chlorophyll breakdown (Akhtar et al. 1999). Studies also found that *gf* was responsible for the retention of chlorophyll in tomato leaves during fruit ripening; its breakdown during leaf senescence was defective. Barry et al. (2008) used a fine-mapping approach to identify a candidate for *gf*. Using positional information from pepper, and the fact that tomato and pepper show a high degree of synteny, a 0.44 cM segment on the long arm of chromosome 8 was identified as the location of *gf*. Bacterial artificial chromosome (BAC) clones from this region were then used

Table 6.2. Tomato fruit color when *green flesh* (*gf*) is combined with various carotenoids and skin colors.

Genotype	Fruit phenotype	Example cultivars
Gf Y R	Red	'Big Beef,' many others
Gf y R	Pink	'Caspian Pink'
Gf – r	Pale yellow	'White Current'
gf Y R	Reddish brown	'Black Prince,' 'Black Cherry'
gf y R	Pinkish purple	'Purple Cherokee'
gf – r	Yellow-green	'Green Zebra,' 'Green Grape'

Y, pigmented epidermis; *y*, colorless epidermis; –, genotype unknown; *R*, red interior; *r*, pale yellow interior.

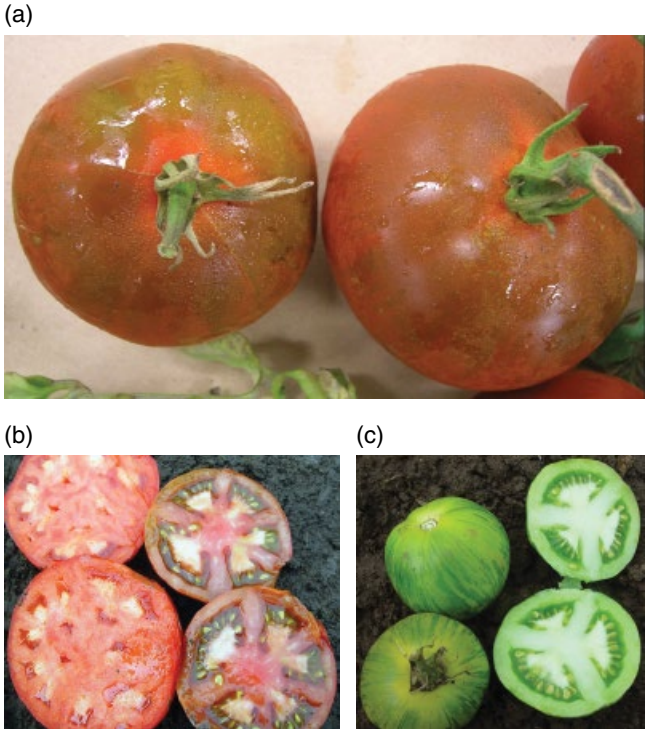

Fig. 6.5. Tomato fruit with the stay-green phenotype conditioned by *green flesh* (*gf*). (a) Mature fruit from the cultivar 'Black Prince,' showing external appearance. Fruit color is affected by *gf* and *Y* (colored epidermis), and by the presence of lycopene. (b) Mature tomato fruit, cut to reveal the interior color: normal red fruit on left, *gf* type on right. (c) Tomato cultivar 'Green Zebra,' combining the carotenoid precursor phytoene with *gf* to produce a "green ripe" fruit color.

to identify a unigene annotated as senescence-inducible *SlSGR1* and corresponding to *OsSGR*. A point mutation causing an amino acid substitution in a highly conserved region of SGR was associated with the altered phenotype of *gf*. In a subsequent study, the *gf* loci in 26 heirloom tomato cultivars were sequenced (Barry and Pandey 2009). Four different null alleles of *gf* were identified among these cultivars, and it was determined that the trait had arisen and been preserved at least five different times.

The estimated global production of tomato in 2014 was over 170.8 million MT (FAOSTAT 2017). The economic value of *gf* is difficult to assess because the trait is most frequently observed in the heirloom types described as "black" and "purple." Overall, the value of all

tomatoes produced in the United States was US$2.1 billion in 2016 (USDA/NASS 2017). The fresh-market portion is valued at US$963 million, but no figures are available on the percentage of the fresh market that is made up of heirloom tomatoes. Our conclusion is that a small but significant fraction of commercial fresh-market production is cultivars that express *gf*, because heirloom specialty types command a premium in the marketplace.

2. Pepper. Pepper is an important vegetable and flavoring crop that is valued worldwide. Five pepper species have been domesticated in the western hemisphere (Bosland and Votava 2000), with *Capsicum annuum* the most widely adapted. Peppers vary substantially, with different fruit colors conditioned primarily by different carotenoid composition.

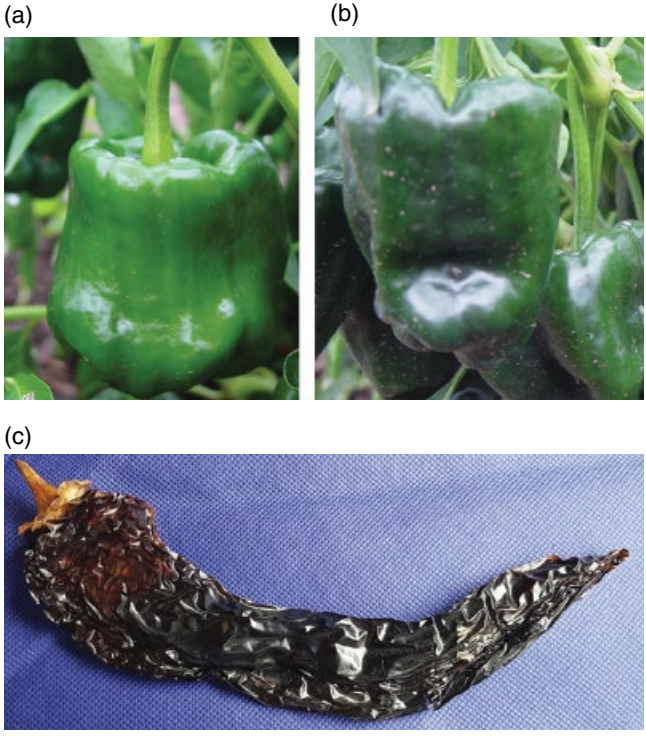

Fig. 6.6. Comparison of normal and *chlorophyll retainer* (*cl*) pepper fruit. (a) Bell pepper, showing light green color. (b) "Ancho"-type pepper, with dark green color conditioned by *cl*. (c) Mature and dried "Pasilla"-type pepper, with *cl*.

The pepper chlorophyll retainer (*cl*) mutation is characterized by inhibition of chlorophyll degradation during fruit ripening (Borovsky and Paran 2008). The ripened fruits of *cl* pepper that contain chlorophyll derivatives and red carotenoids are brown (Fig. 6.6), while those that contain chlorophyll derivatives and yellow carotenoids are green. Borovsky and Paran (2008) found that the *cl* mutation inhibits chlorophyll degradation during fruit ripening and leaf senescence. Roca and Mínguez-Mosquera (2006) characterized chlorophyll catabolism in normal and *cl* pepper lines. They found that the ratio of chlorophyll a to chlorophyll b shifted in *cl* but not wild-type as senescence progressed, and concluded that a change associated with PaO activity was responsible for the *cl* genotype. In another study, Roca et al. (2006) found that *cl* was associated with a slower rate of carotenoid synthesis and accumulation. Carotenogenesis was independent of chlorophyll catabolism, but the two processes interacted during the conversion of chloroplasts to chromoplasts. Efrati et al. (2005) mapped *cl* to chromosome 1 in pepper using a BC_1 interspecific (*C. annuum* × *Capsicum chinense*) mapping population and showed that this location was homologous to chromosome 8 in tomato. They also mapped three loci of *CHLASE* and one locus of *PaO* in order to determine whether these co-segregated with *cl*, but found that they mapped to different chromosomes. The pepper stay-green (*CaSGR*) sequence has been isolated and co-segregated with *cl* in a BC_1 mapping population (Borovsky and Paran 2008). Borovsky and Paran (2008) sequenced the wild-type and mutant alleles, and found tryptophan was substituted for arginine at position 114 in the protein sequence. The single-nucleotide polymorphism (SNP) that differentiates the wild-type and mutant alleles was exploited to develop a polymerase chain reaction (PCR) marker useful for marker-assisted selection (Borovsky and Paran 2008).

Using semiquantitative reverse transcriptase (RT)-PCR, Borovsky and Paran (2008) studied the expression of *CaSGR* and found it to be induced during leaf senescence and less so during fruit ripening. Analysis of expression data and the genetic sequence suggests that *CaSGR* regulates the *cl* mutation in pepper (Kusaba et al. 2007).

The chlorophyll retainer trait is found commercially in Poblano or Ancho and in Chilaca or Pasilla (*C. annuum*) pepper types. The trait appears to be established in the *C. chinense* gene pool as well, but homology between the genes in *C. annuum* and *C. chinense* has not been established. These make up a minor market class in the United States, but are widely used throughout Latin America. Worldwide production of dry and green peppers in 2014 was 3.8 and 32.3 million MT, respectively (FAOSTAT 2017). In the United States, chili peppers were

valued at US$162.9 million in 2016 (USDA/NASS 2017). It is unknown what proportion of these carry the *cl* mutation.

D. Rutaceae

1. Citrus. Alós et al. (2008) were the first to identify *navel negra* (*nan*), a stay-green mutation in citrus that imparts a distinctive fruit phenotype. Their analysis of *nan* at the molecular level and their study of metabolite data suggest that it represents a new class of stay-green mutation. The *nan* mutation may result from a lesion in a regulatory pathway upstream of the signal-inducing chlorophyll degradation. It exhibits symptoms of oxidative stress prior to the onset of the normal senescence process.

IV. OTHER CROPS THAT MAY CARRY COSMETIC STAY-GREEN MUTATIONS

Several crops show phenotypic characteristics of stay-green traits, but have not yet been characterized at the molecular level to confirm a mutation in the chlorophyll senescence pathway. These crops include cucumber, lentil, lima bean, mung bean, cowpea, and fava bean, among others (Table 6.3). In this section, we describe what is known about the potential stay-green trait in these crops.

A. Fabaceae

1. Lentil. An important food source, especially in the Middle East and India, lentil is produced globally. It has several different seed types that vary in color from red to orange, yellow, and green.

Researchers (Singh 1978; Slinkard 1978) have investigated the segregation of orange or red cotyledon color versus yellow color, and

Table 6.3. Crops with economically important potential stay-green variants.

Crop	Scientific name	Trait
Chickpea	*Cicer arientium*	Seed coat/cotyledon color
Cowpea	*Vigna unguiculata*	Seed coat/cotyledon color
Cucumber	*Cucumis sativus*	Fruit color
Fava bean	*Vicia faba*	Seed coat/cotyledon color
Habañero pepper	*Capsicum chinense*	Fruit color
Lentil	*Lens culinaris*	Seed coat/cotyledon color
Lima bean	*Phaseolus lunatus*	Seed coat/cotyledon color
Mung bean	*Vigna radiata*	Seed coat/cotyledon color

found that red is controlled by a single dominant gene. More recently, researchers have recognized that the pigment in lentil cotyledons is regulated by three genes: *Dg*, *Y*, and *B*. Dominant *Y* and *B* produce yellow and brown pigments, respectively. In the recessive state (*dg dg*), no pigment is produced and a dark-green color is perceived. If *Dg* is dominant but both Y and *B* are recessive (*Dg-yybb*), no pigment will be produced and cotyledons will be light-green (Emami and Sharma 1996a, b). Sharma and Emami (2002) indicated that *Dg* is epistatic to *Y* and *B* simultaneously, because the *Dg* process begins earlier in the metabolic pathway of pigment synthesis than does the process of either *Y* or *B*. Lentil and pea show a high degree of synteny (Moffet 2006), and *SGR* in lentil has been mapped to an orthologous linkage group to chromosome 1 in pea containing the *I* gene (N. Weeden, University of Montana, pers. comm.). It is not known whether *SGR* in lentil co-segregates with *Dg*, *Y*, or *B*.

2. Cowpea, Blackeye Pea, and Southern Pea. *Vigna unguiculata* subsp. *unguiculata* is referred to as "cowpea" through much of the world and is mainly used as a dry edible grain. Adapted to warmer climates with limited moisture, it was first domesticated in Africa before diffusing across the globe (Xiong et al. 2016).

In the southeastern United States, where it is grown commercially to be canned or frozen, cowpea may be referred to as "southern pea" or "blackeye pea." Until the 1990s, southern peas had cream cotyledons. Separate genes for persistent green cotyledon (*gc*) and green testa (*gt*) were discovered and were determined to be non-allelic. Fery and Dukes (1994) developed the cultivar 'Bettergreen,' which possesses *gc*. The *gt* gene that conditions green testa was incorporated into the cultivar 'Freezegreen'. Fery and Dukes (1994) found that by combining these two genes, they could give seed a higher and more uniform color quality than that harvested from either 'Bettergreen' or 'Freezegreen' alone. Contemporary cultivars have both the *gc* and *gt* genes, and produce seed with an enhanced and persistent green color and distinctive and darker green foliage (Fery and Thies 2000; Fery 2005, 2006, 2009). However, the retention of green foliage during senescence has not been reported. One or both genes are candidates for *SGR*.

Production figures for southern pea in the southeast U.S. are difficult to come by, but were estimated in the 1990s to be about 11 000 ha grown for frozen and canned southern peas, and another 30 000 ha planted in home gardens for fresh southern peas (Gómez 2004). With yields of 1.95 $Mt \cdot ha^{-1}$ and price of US$0.66 kg^{-1} (Gómez 2004), total value would be about US$52 million per year. Approximately 50%

of cultivars used for commercial production in the southeastern U.S. have the combined *gc* and *gt* genes (R. Fery, USDA-Charleston [retired], personal communication).

3. Mung Bean. *Vigna radiata* is a domesticate of northwestern India (Fuller 2007), and is widely used throughout the region as whole beans, paste, sprouts, and a source of starch for noodles (Tang et al. 2014). The majority of mung bean cultivars have yellow cotyledons. Thakare et al. (1980) found an induced mutation for green cotyledon in mung bean that behaved as a monogenic recessive. Like cowpea, it is designated *gc* for green cotyledon, and may be a candidate for *SGR*. Whether the original germplasm has been incorporated into commercial cultivars, or even remains extant, is unknown.

4. Fava Bean. *Vicia faba* represents the first crop for which the term "stay-green" was used to describe persistent green foliage (Sjödin 1971; Thomas and Ougham 2014). Domesticated in the Middle East, it is widely grown as a winter annual in hotter climates and as a summer annual at higher latitudes.

Both dry seed and mature, high-moisture seed are consumed. Two genes for green cotyledon color (*il-1*) and green testa color (*y*) are known, while the wild-type produces yellow cotyledons and a buff-colored testa. Sjödin (1971) identified a number of seed color variants, some of which possess a green seed coat. Five accessions (numbers y-1 through y-5) have been identified with green seed coat. Among these, y-4, obtained from Bond in 1961, is identified as being "stay-green." Duc et al. (1999) discovered *il-1* as a spontaneous mutation and determined that it was inherited as a single recessive gene that was not allelic to *y*. Khazaei et al. (2014) mapped a green seed-coat trait to linkage group 6 (chromosome IV) and designated the locus as *Sc*. Apart from Duc et al. (1999), fava bean researchers have not distinguished between green testa and green cotyledons, making it difficult to determine whether one or more of the genes discovered might be a candidate for *SGR*. In any case, germplasm with the trait appears to be available for researchers interested in identifying *SGR* variants in fava bean.

5. Lima Bean. Although crossing between the species has not produced fertile offspring, *Phaseolus lunatus* is similar genetically to *P. vulgaris*, including in the fact that it originated in two centers of domestication. Large-seeded 'Big Lima' types descend from the Andean gene pool, whereas small-seed 'Sieva,' 'Potato,' and 'baby' types are of Mesoamerican origin. Within the baby lima class, a green

cotyledon trait similar to *pc* in common bean was identified and incorporated into the market class (Magruder and Wester 1941). The trait has been utilized in modern baby lima cultivars, such that most nowadays possess it (Wester 1965). For this trait, the stay-green appearance of the vegetation during senescence has not been reported and the pleiotropic effects observed with common bean (white cotyledons upon germination, poor emergence) have not been observed.

B. Cucurbitaceae

1. Cucumber. Many heirloom and landrace cucumbers (*Cucumis sativus*) and their wild relatives have fruit with black spines that turn to an orange or yellow color as they mature (Fig. 6.7a). Contemporary commercial cultivars may have white spines on fruit that remain green at maturity (Fig. 6.7b). This trait has been widely incorporated; it is preferred in both slicing and pickling cucumbers, because the fruit retain a more uniform color for a longer period. Spine color is controlled by four independent genes. The black spine trait conditioned by *B* (Xie and Wehner 2001) is closely associated with orange (*R*) mature fruit color and increased netting (*H*). The recessive *b-1* allele produces white-spined fruit that remain green at maturation and lack netting. A differential response in foliage between the two alleles has not been reported.

Li et al. (2013) mapped the *B* locus to the short arm of chromosome 4 and used a fine-mapping approach to propose a candidate gene for

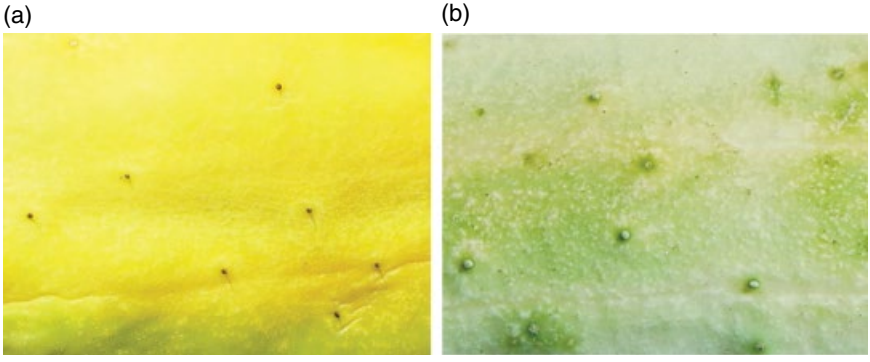

Fig. 6.7. Close-up of the epidermis and spines of senescing cucumber fruit. (a) Normal epidermis, showing yellow color with black spines at maturity, conditioned by *B*. (b) Green skin and white spines, conditioned by *b* (green epidermis also exhibits photobleaching).

this trait. Their candidate is an *R2R3-MYB* transcription factor, which was one of six gene models found within the circumscribed region. They speculate that this *R2R3-MYB* controls anthocyanin expression in the spines. However, this hypothesis is suspect, as it is not apparent that any of the Cucurbitaceae produce anthocyanins among the rather low quantities of flavonoids that they do produce (Bhagwat and Haytowitz 2015). Validation by transformation to verify phenotypic expression of this candidate has yet to be performed. As an alternative hypothesis, the *R2R3-MYB* could control melanin metabolism. Melanins appear as black or brown pigments (Solano 2014), which more closely resemble the spine color in *Cucumis* spp. than would anthocyanin accumulation. In plants, melanin biosynthesis occurs through various pathways, but these have not been well characterized, and it has not been determined where biosynthesis occurs. If melanin biosynthesis and regulation are involved in the black spine trait, then this may represent a hitherto unknown interaction with chlorophyll catabolism.

V. UTILITY AND IMPORTANCE OF COSMETIC STAY-GREEN

Cosmetic stay-green mutations contribute to an increased quality of a number of crops, including dry edible peas and beans, garden peas and snap beans, tomatoes, and peppers. In some cases (e.g., peas), the economic value of cosmetic stay-green is considerable; in others, cultivars with stay-green constitute a minor part of crop usage. Overall, the value of the stay-green characteristic has a positive economic impact all along the supply chain, from the producer to fresh markets and processing industries, and has potential to benefit other crops.

Stay-green would appear to have utility in many fruit and vegetable crops where it has not been discovered. In particular, it might have useful properties in the leafy green vegetables in the families Asteraceae, Chenopodiaceae, and Brassicaceae (Table 6.4). Any crop where extended shelf-life is important could be a candidate for incorporation of cosmetic stay-green.

SGR mutations may prevent chlorosis symptoms induced by pathogens. This was found for an *SGR* in *Arabidopsis* in the presence of *Pseudomonas syringae* and *Alternaria brassicicola* (Mecey et al. 2011). Although it is unclear whether reduction in this disease symptom has other benefits to the host organism, this work points to the need to investigate whether stay-green mutations can reduce disease in other crops, especially those of economic importance.

Table 6.4. Specialty crops where cosmetic stay-green might be useful.

Crop	Scientific name	Trait
Alfalfa	*Medicago sativa*	Leaf greenness, protein content, forage quality
Broccoli	*Brassica oleracea*	Green color retention in head during storage
Cabbage	*Brassica oleracea*	Color retention in head during storage
Kale	*Brassica oleracea*, *B. napus*	Leaf greenness
Lettuce	*Lactuca sativa*	Leaf greenness
Melon	*Cucumis melo*	Disease resistance
Spinach	*Spinacia oleracea*	Leaf greenness
Swiss chard	*Beta vulgaris*	Novel leaf colors

Qian et al. (2016) found that a nonfunctional allele of *NYE1* increased chlorophyll content and improved oil yield in oilseed *Brassica napus*. This allele was introgressed from *Brassica rapa* into *B. napus*, apparently through unconscious selection by plant breeders. Other oilseed crops might benefit from the incorporation of cosmetic stay-green, but researchers would need to be cognizant of the "green seed problem" (see later).

Beyond food crops, cosmetic stay-green may have utility for turf grasses (Thorogood 2003), as well as in the ornamental industry, where retention of green color after crop harvest is desirable (van Geest et al. 2017).

VI. LIMITATIONS OF STAY-GREEN

In most cases, the principle influence of the stay-green trait is on the color of fruits, seeds, and foliage, caused by retention of green chlorophyll catabolites. In some crops (primarily grain/vegetable legumes), the stay-green trait may be associated with undesirable effects. In addition to distinctive green color during vegetative and reproductive growth and development and delayed senescence, bean seed of *pc* plants has white cotyledon color upon germination (Fig. 6.4d). Rates of germination and emergence are reduced in these white cotyledon types compared to non-*pc* seed (Fig. 6.8). The reduced germination of *pc* types is observable in data from segregating F_2 populations (Table 6.5). Overall, a roughly 20% reduction has been observed in the homozygous *pc* category compared to what was expected. It is not known whether white cotyledon color has other deleterious effects on growth, but if cotyledons are photosynthetically active upon emergence, then photosynthesis in *pc* types may be reduced.

Of greater concern is the reduction in germination associated with *pc* in bean. The effect can be mitigated by treating the seed with fungicides

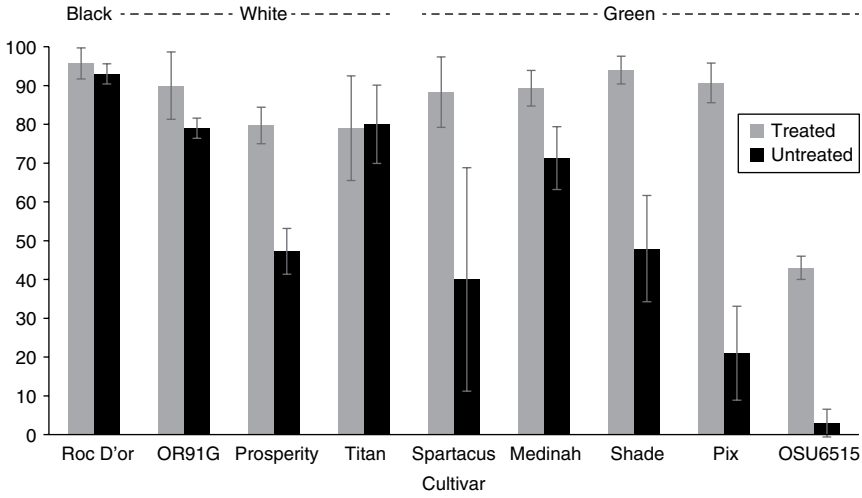

Fig. 6.8. Differences in germination of snap bean lines carrying different seed testa colors when planted with or without seed-applied fungicide in a trial performed at the OSU Vegetable Research Farm in 2012. (Figure modified from Al-Jadi et al. 2016). The black-seeded cultivar is *PP PcPc*, the white-seeded cultivars are *pp PcPc*, and the green-seeded cultivars are *pp pcpc*. (With *P-*, other color genes are expressed; with *pp*, color is suppressed and seeds and flowers are white.)

Table 6.5. Segregation for *persistent color* (*pc*) in nine experimental green bean populations, tested for fit to a 3 : 1 ratio.

Cross	*Pc-*	*pcpc*	Total	χ^2	Prob.
Goldrush × I-201	952	290	1242	1.8	0.179
Goldrush × I-207	458	137	595	1.2	0.266
Goldrush × 8033	160	28	188	10.2	0.001
Goldrush × 8036	296	75	371	4.5	0.033
Earliwax × I-201	497	120	617	10.1	0.001
Earliwax × I-207	736	159	895	25.0	0.000
Earliwax × 8033	108	32	140	0.3	0.558
Earliwax × 8036	332	71	403	11.7	0.001
Earliwax × I-207	416	77	493	23.1	0.000
Total	3955	989	4944	65.8	0.000
Homogeneity				22.3	0.004

Data from a study of the interaction between *pc* and *y*, the wax bean trait. See Baggett et al. (1999) for details on parents used in the crosses. A cumulative deficit of homozygous *pc* individuals across crosses is apparent.

prior to planting (Fig. 6.8). This suggests that *pc* types may be more susceptible to soil-borne fungi, but whether *pc* types lack antifungal compounds normally found in non-*pc* seed or if *pc* seeds exude greater volumes of compounds into the rhizosphere, attracting pathogens, is unknown. Seed with *pc* may also have thinner testas and be more fragile, thereby increasing imbibitional cracking and providing entry points for pathogens. A substantial portion of the snap bean acreage in the United States is planted to *pc*-bearing cultivars, although the use of this trait may be declining as breeders are finding other genes that produce an attractive pod color and are not associated with germination problems.

In dry peas, a slight reduction in the germination of green cotyledon types compared to yellow cotyledon types has been observed (K. McPhee, Montana State University, pers. comm.), but the effect is subtle. There are anecdotal reports that when green cotyledon peas were first observed in the Middle Ages, their germination was substantially worse than that of yellow cotyledon peas. It may be that over time, the selection and breeding of green cotyledon peas have led to the development of cultivars that compensate for reduced germination effects.

Environmentally induced cotyledon bleaching of green cotyledon types has been found to be a problem in lima bean and pea (Wester and Jorgensen 1956; Pollock and Toole 1964; Maguire et al. 1973). Unlike in common bean, where cotyledons of germinating seeds are inherently white, bleaching seems to be associated with weathering of the seed by sun and rain after the plant senesces but before the seed can be harvested. When bleached and unbleached seed lots were compared, the bleached seeds had higher levels of electrical conductivity, were leaking more soluble solids into the surrounding media, and had a reduced level of germination (Maguire et al. 1973). Those seedlings that did germinate had reduced vigor relative to seedlings from unbleached seed lots. In lima beans, similar results are observed, but there also appears to be a genetic component, whereby some green-seeded cultivars are more prone to bleaching than others (Wester and Jorgensen 1956; Pollock and Toole 1964). In some *pc* cultivars of snap bean, we have observed mixed cotyledon colors during field emergence. These may be the result of mixes with non-*pc* cultivars, or they may represent natural genetic variation for susceptibility to bleaching; perhaps recurrent selection could be implemented for green cotyledon color in snap bean in order to develop *pc* types with improved germination.

In grain and oilseed crops of soybean and rapeseed (*B. napus*), green seed is undesirable because it interferes with oil quality and may reduce

seed viability (Smolikova et al. 2017). Retention of photoreactive chlorophyll and its pigmented derivatives may produce reactive oxygen species, which increase rancidity in oil and have a noticeable effect on taste and smell. For rapeseed, spring frosts during seed maturation will inactivate enzymes in the chlorophyll catabolic pathway, resulting in the accumulation of chlorophyll and derivatives. For soybeans, drought and high temperatures can impair SGR and NYC1 enzyme activity, leading to the accumulation of chlorophyll in the seed, which decreases nutritive value and affects quality.

In soybean, Luquez and Guiamét (2002) found that cosmetic stay-green types have higher levels of stomatal conductance than normal soybeans as senescence progresses. They concluded that these stay-green types are more sensitive to drought stress. The association between stay-green traits and drought stress in other crops has not been investigated.

Another concern with type C stay-green plants is that the interrupted chlorophyll catabolic process does not recycle nitrogen, metal ions, and other elements useful for plant growth and development, which may reduce yields. Studies in perennial ryegrass and rice have not found a negative effect on yields due to reduced N recycling in stay-green genotypes (Thomas and Ougham 2014). It has been hypothesized that species with weak sink activity will not suffer from the lack of remobilization, but that those crops with high sink demands will have a yield penalty.

VII. PROSPECTS AND THE NEED FOR FURTHER RESEARCH

Understanding the basis for the onset of senescence in plants and the isolation of the genes involved in the senescence process is of fundamental importance for crop improvement. This field of study has been active mostly within the last 20 years, but it is mainly in the past decade that so much has been accomplished in our understanding of the senescence process. This is primarily due to the deployment of genomics and bioinformatics tools and the sequencing of major crop genomes. For a trait such as cosmetic stay-green, where genes in the chlorophyll catabolic pathway are highly conserved, ample opportunity remains to extend these tools to other crops where little work has been conducted on this subject. There is a need to verify stay-green in a number of crops that appear to have likely candidates for the trait (Table 6.3). There also is a need to identify novel stay-green traits in other crops that could benefit from their incorporation (Table 6.4). These would include green vegetables, cut flowers and other ornamentals, and forages. For example,

broccoli de-greening in storage is an important problem, and *SGR* plays a role in this process. Cyokinin treatments inhibit *SGR* activity and result in prolonged greenness, whereas ethylene enhances *SGR* activity and causes more rapid yellowing (Gómez-Lobato et al. 2014). To date, no stay-green broccoli cultivars have been identified, but such a trait could be very useful in this crop.

Stay-green may be particularly useful for forage crops. An *SGR* from *Medicago truncatula* was used to develop an RNAi construct that was transformed into alfalfa (*Medicago sativa*). The dried forage crop showed improved green appearance that persisted much longer than in normal alfalfa. It also had 50% more chlorophyll and an increased crude protein content (Zhou et al. 2011).

The potential effects on human health of the byproducts of altered chlorophyll degradation have not been addressed in any depth. Chlorophyll has antioxidant properties, but some intermediates in the catabolic process are strong oxidants. In eating ripe fruits and vegetables, we routinely ingest the catabolic end-products that are colorless type-I and type-II phyllobilins (Kräutler 2016). The nutritional effects of these compounds are not well known, but they have been shown to be strong antioxidants in their own right (Kräutler 2008). Research in this area presents an opportunity to gain new knowledge on human health and nutrition.

In vegetables, dark green color is associated with higher levels of carotenoids; this is another property for investigation in stay-green types. In snap beans, *pc* types are among those with the greenest pods and the highest carotenoid levels (unpublished data). In both tomatoes and peppers, altered accumulation of carotenoids has been found associated with the prolonged transition period from chloroplast to chromoplast (Roca et al. 2006; Barry 2009). Altered accumulation of carotenoids may have implications for taste and flavor, as well as nutrition, both because carotenoids have direct health benefits and because carotenoid derivatives may affect taste and aroma (Lira et al. 2016). We have observed that the "purple" and "black" heirloom tomatoes almost always have excellent taste, flavor, and aroma. It is not clear whether this is an effect of the *gf* locus or a byproduct of being an heirloom.

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